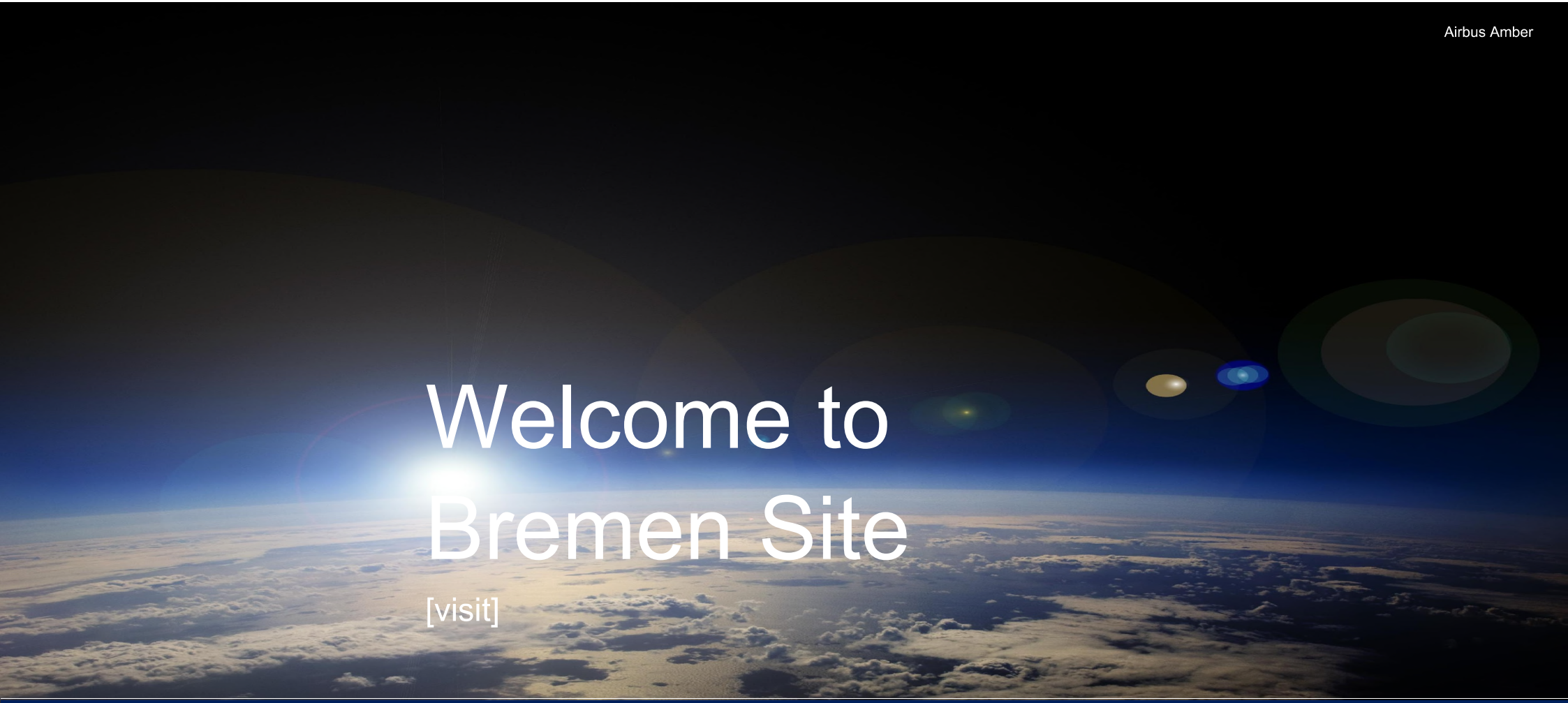


Welcome to Bremen Site

[visit]

[name, job title]

AIRBUS



Bremen: An Airbus Ecosystem

Airbus Operations

~ 2100 employees

Engineering: materials research and testing, flight physics, high-lift, cabin & cargo
Manufacturing: high-lift assembly, A330 & A350 wing equipping
Maintenance & Services: European Repair Station
Other: controlling, accounting, procurement for aerostructures, one vocational training for the site

Airbus Aerostructures

~ 350 employees

Competence centre for development and production of **metal and thermoplastic forming parts** produced in a fully integrated end-to-end process flow

Testia

~ 80 employees

Non-destructive testing services, products, engineering and training; **destructive testing**; sensor technology and data analytics

Airbus Defence and Space

~ 1000 employees

A400M: development & integrated fuselage assembly, transport & delivery capabilities,
Space exploration: Orion ESM development & production, technology development for lunar exploration, low observability & systems, ISS operations and launch of payloads

Airbus DS Airborne Solutions

~ 280 employees

Unmanned Aircraft Systems: systems engineering (integration, design, production, maintenance), permits for operations and services, mission management
Cargo Loading Systems: equipment design, production and maintenance/repair/overhaul

ArianeGroup

~ 600 employees

Development and manufacturing of **Ariane launcher**‘s upper stage, future upper stages and propellant tanks

AIRBUS

Our purpose

We pioneer sustainable aerospace for a safe and united world

R&T and Engineering focus on:

- Light-weight technologies
- Modular & scalable platforms
- Life-cycle analysis and management
- Renewable materials
- Low-energy technologies
- End-to-end simulation
- Repair services and recycling

Manufacturing focus on:

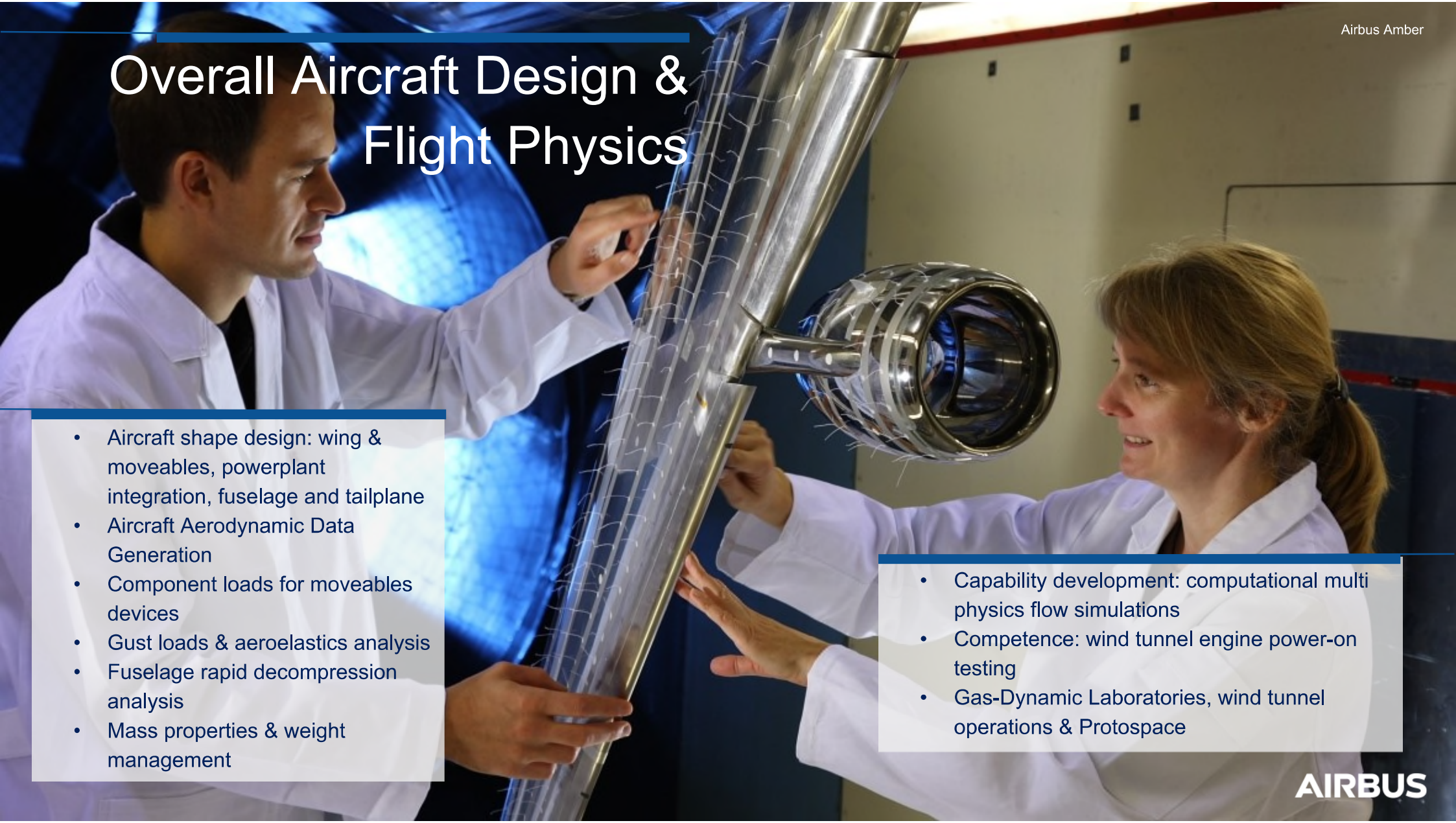
- Low facility energy consumption
- Flow optimisation
- Logistics optimisation

AIRBUS

Airbus Operations

Engineering and Manufacturing Activities

AIRBUS



Overall Aircraft Design & Flight Physics

Airbus Amber

- Aircraft shape design: wing & moveables, powerplant integration, fuselage and tailplane
- Aircraft Aerodynamic Data Generation
- Component loads for moveables devices
- Gust loads & aeroelastics analysis
- Fuselage rapid decompression analysis
- Mass properties & weight management

- Capability development: computational multi physics flow simulations
- Competence: wind tunnel engine power-on testing
- Gas-Dynamic Laboratories, wind tunnel operations & Protospace

AIRBUS

Structural and System Engineering

Wing high-lift structure & systems

- Design of high-lift moveable structures and systems
- Structure & system ground and flight testing
- Development and certification of future high-lift moveable technologies

Aircraft hydraulic system

- Definition, development and certification of hydraulic power system (i.e. flight controls, high-lift system, landing gears, thrust reverser, cargo door actuation)



Structural design & system integration

- Definition, development and certification of fuselage & door structure for all commercial aircraft
- Mechanical and electrical system integration
- Structural repair services

Cabin & cargo systems

- Definition, development and certification of cargo compartments and cabin safety systems
- Testing of compartments, systems and components
- Customization of products and solutions based on customer expectations

AIRBUS

Flight and Integration Test Centre



System integration test

- From full scale "Zero Means" ...
- … over control computer testing in fully simulated environment ...
- ... to interconnected testing with test benches on other Airbus sites

- Testing of high-lift devices, doors, hydraulics
- R&T and virtual testing
- On-site telemetry
- preparations for hydrogen test activities

High-Lift Assembly

- Carbon fiber and aluminum flaps for all commercial aircraft and for the A400M
- 11 production lines with a high degree of automated processes



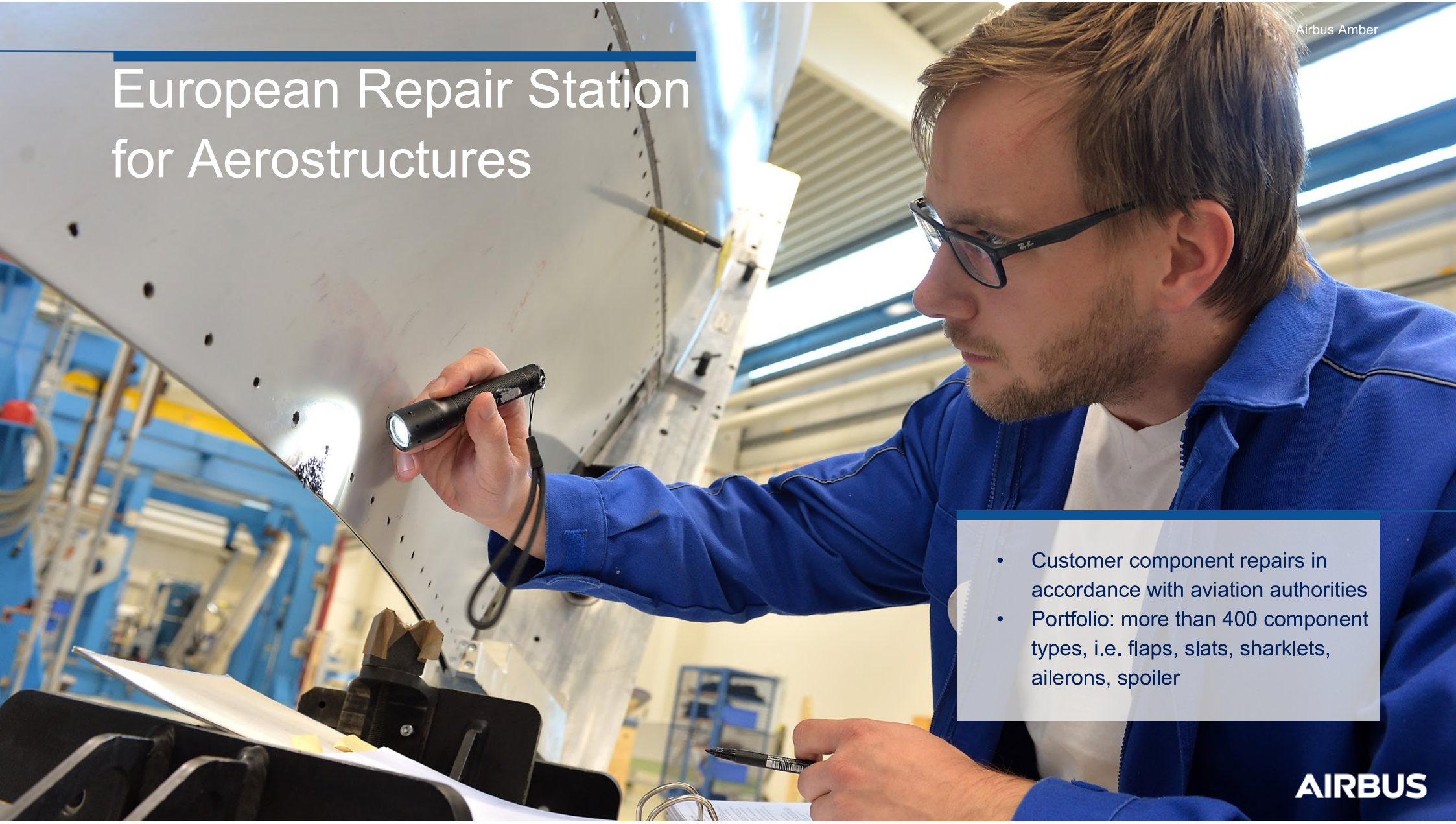
Wing Equipping



- Equipping of aluminium-alloy and composite A330 and A350 wings incl. hydraulic, electrical and mechanical systems
- Functional test of all movable parts
- Customer inspection: on-site support

AIRBUS

European Repair Station for Aerostructures



- Customer component repairs in accordance with aviation authorities
- Portfolio: more than 400 component types, i.e. flaps, slats, sharklets, ailerons, spoiler

AIRBUS

Airbus Amber

Strategic initiatives



OpenCargoLab

- Initiative to improve digitalisation, sustainability and safety in the cargo industry and strengthen the Airbus position in the air freighter market

Moveables Competence Centre

- Initiative to unite Bremen's high-lift expertise in order to redesign, assemble and repair high-lift systems more efficiently in the future.

AIRBUS

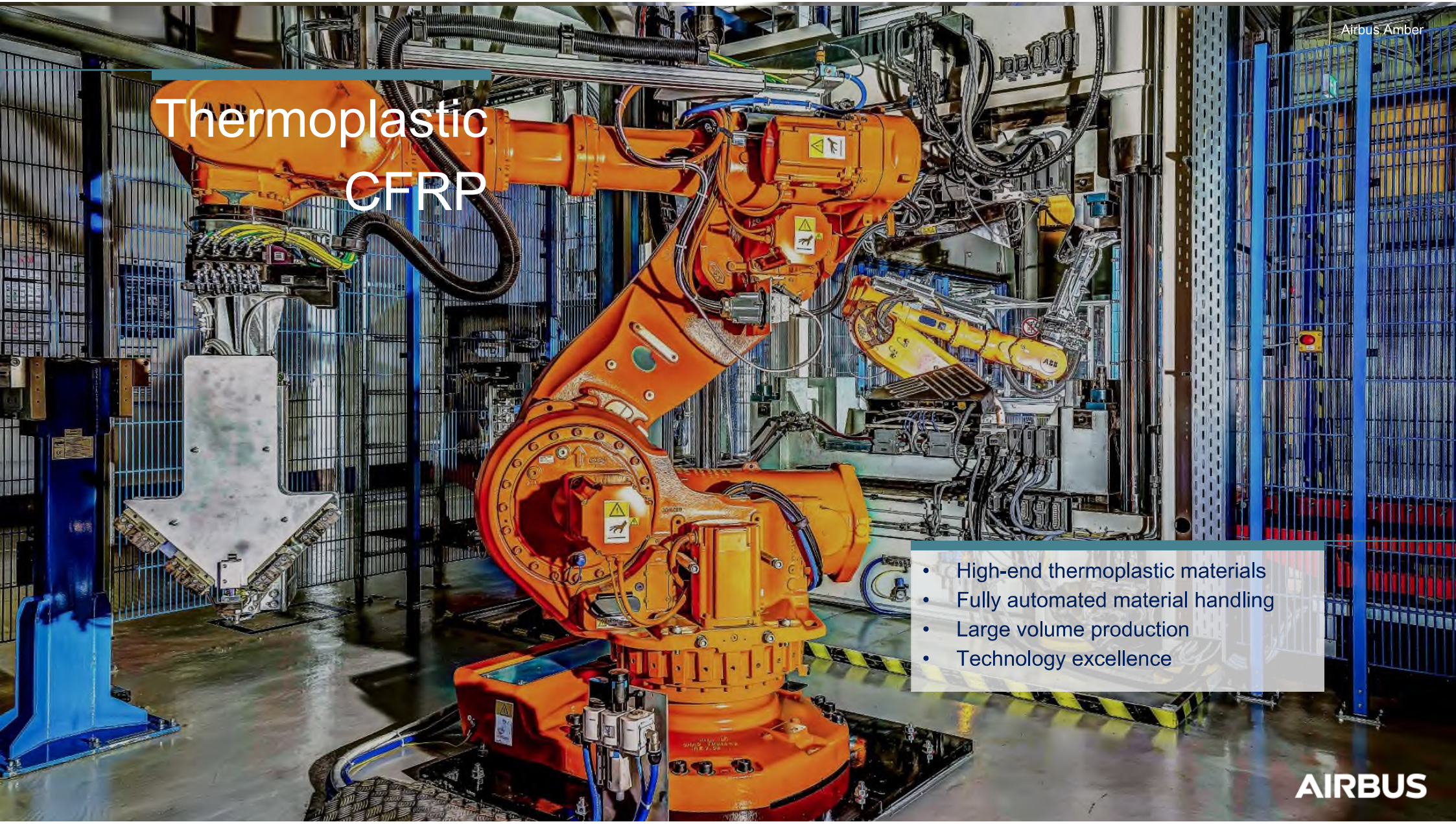
Airbus Aerostructures

AIRBUS

Parts production



- Development and production of metal and thermoplastic forming parts
- 3,5 million aluminum, titanium, steel and CFRP parts per year
- Supplier for all Airbus commercial and military aircraft
- Full end-to-end and digital production environment
- Speedline delivery within 48 hours



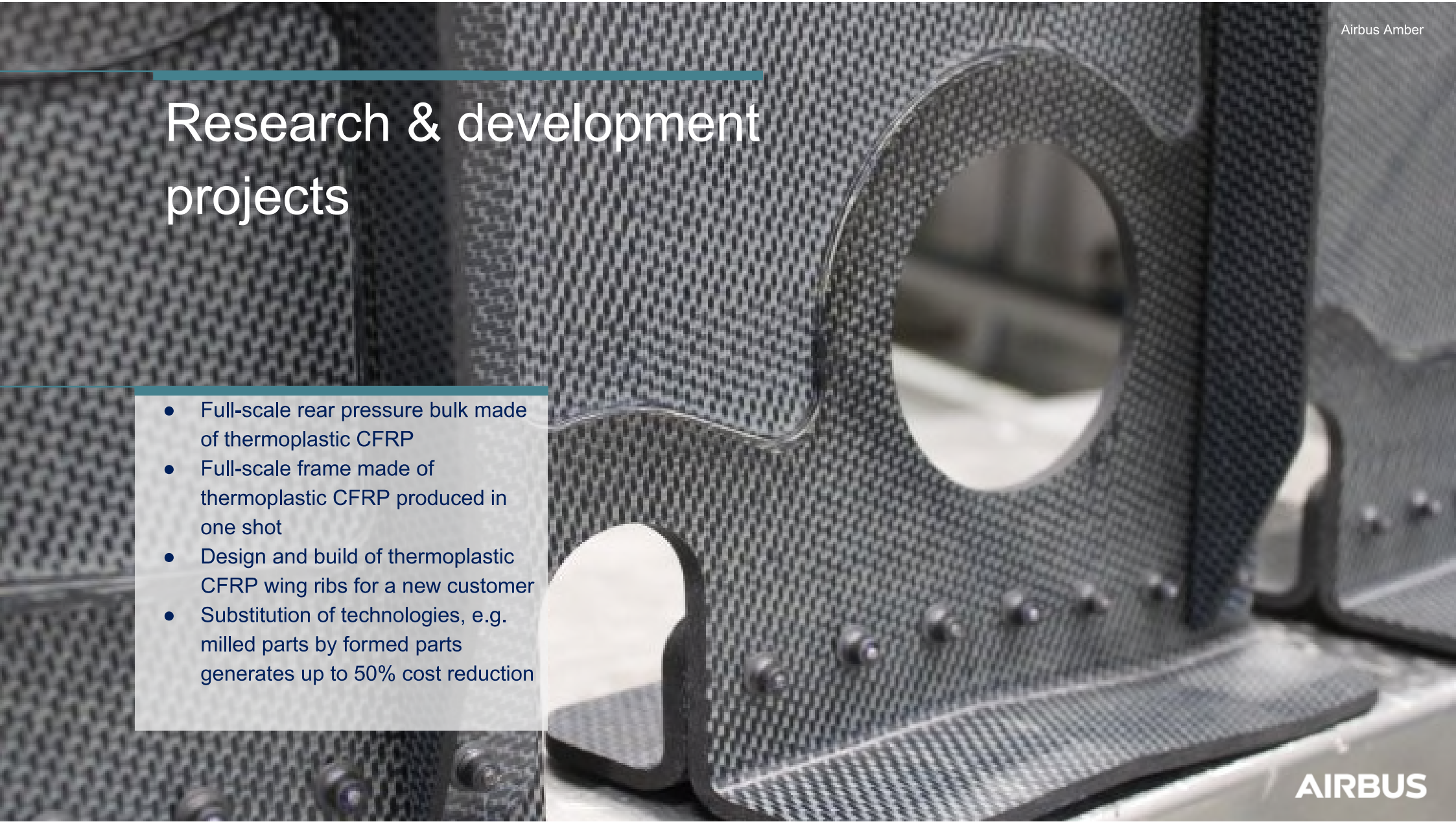
Thermoplastic CFRP

- High-end thermoplastic materials
- Fully automated material handling
- Large volume production
- Technology excellence

AIRBUS

Research & development projects

- Full-scale rear pressure bulk made of thermoplastic CFRP
- Full-scale frame made of thermoplastic CFRP produced in one shot
- Design and build of thermoplastic CFRP wing ribs for a new customer
- Substitution of technologies, e.g. milled parts by formed parts generates up to 50% cost reduction

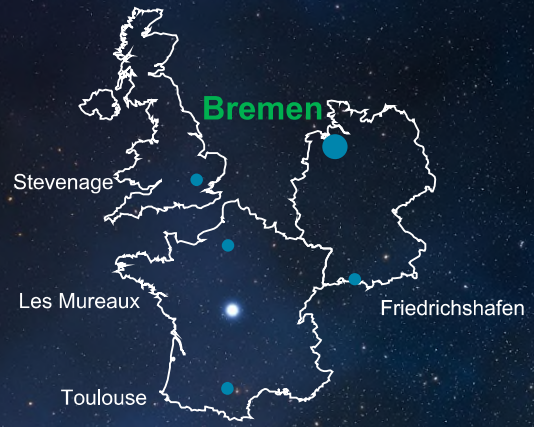


Airbus Defence and Space

Space Systems

AIRBUS

Airbus Space Exploration:
From Bremen to the Moon (and beyond)





EXPANSION



UTILIZATION



DISCOVERY

3 DESTINATIONs **3** PURPOSES

AIRBUS

Orion European Service Module

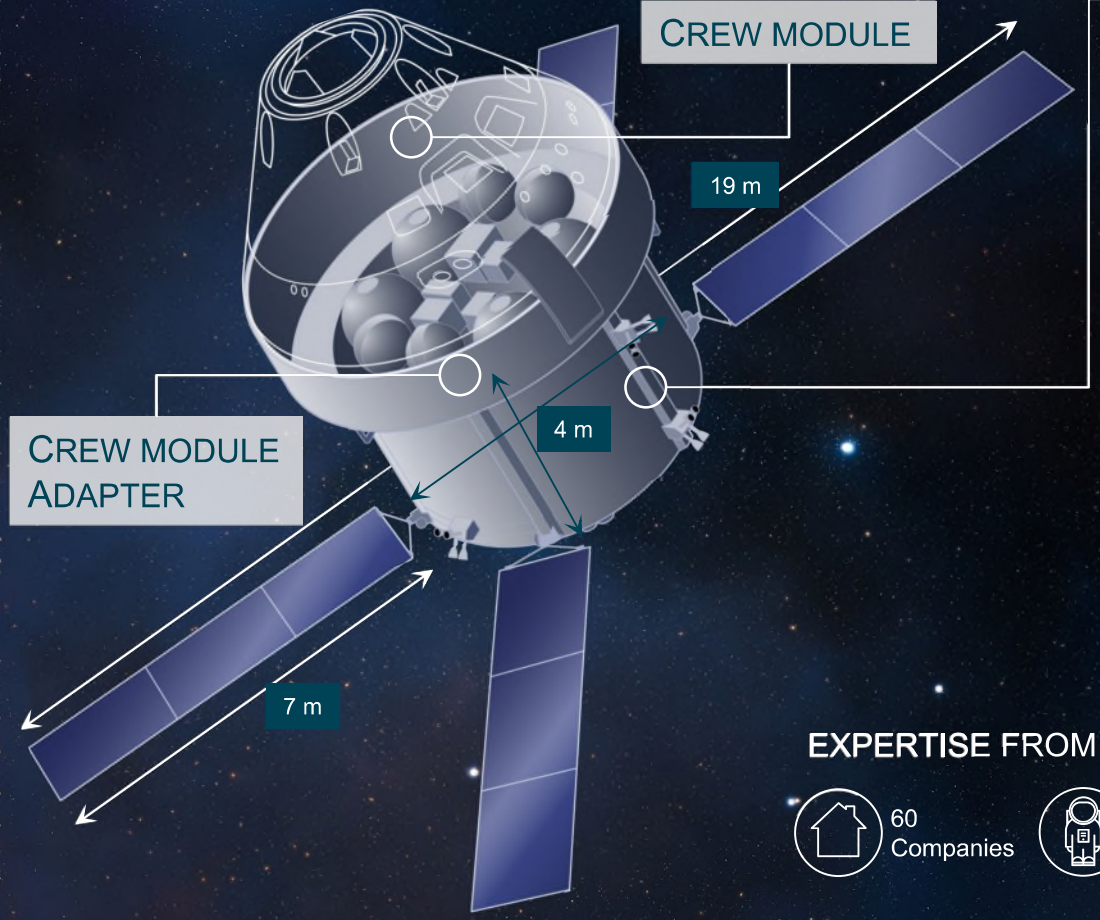


- Airbus is ESA‘s prime contractor for the European Service Module of NASA‘s new spaceship Orion.
- Taking humans back to the Moon for the first time in 50 years.
- First launch to the Moon: November 2022.
- The ESM is developed and assembled in Bremen.

© NASA 2022

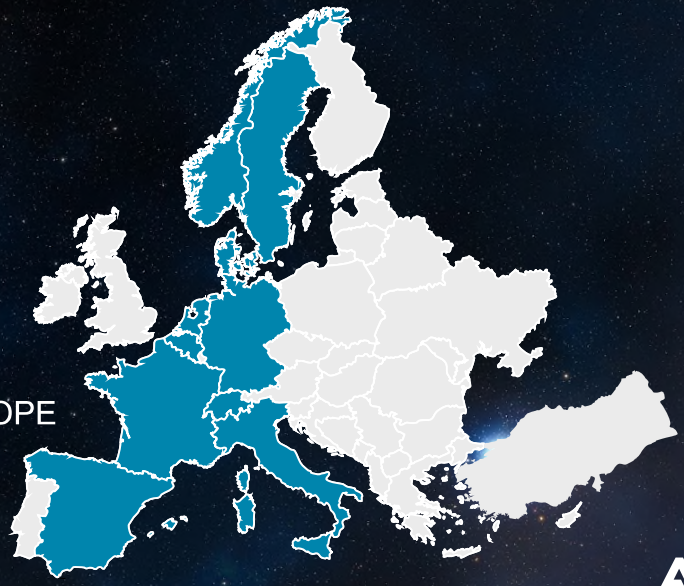
AIRBUS

Made in Bremen: The European Service Module (ESM)



SERVICE MODULE ESM

- PROPULSION
- ENERGY
- CONSUMABLE (OXYGEN & WATER)
- THERMAL CONTROL



EXPERTISE FROM ACROSS EUROPE

60 Companies

3.000+ Employees

AIRBUS

Starlab



- A transatlantic joint venture between Voyager Space, Airbus and other international partners that is designing, building, and will operate the Starlab commercial space station.
- Starlab will serve a global customer base of space agencies, researchers and companies, ensuring a continued human presence in low-Earth orbit and a transition of research from the International Space Station into the new commercial space station era.
- Bremen's contribution:
  - Significant part of programme management and (chief-) engineering activities.

Airbus Amber

AIRBUS

International Space Station (ISS)



Airbus is Europe‘s no. 1 company in human spaceflight, providing Europe’s contribution to the ISS:

- Operation of the **Columbus** laboratory, launched 2008, built in Torino and Bremen.
- Development and launch of scientific payloads used for microgravity research and new technologies.
- Bremen contributes within the Columbus utilization service with engineering support in flight- and ground segment.

AIRBUS

Microgravity Science: Sounding Rockets



- Sounding rockets TEXUS/MAXUS for various experiments, e.g. in biology, medicine, life science, fluid physics, combustion, material science, technology demonstration.
- Airbus operates the TEXUS/MAXUS programme to provide a microgravity environment of up to 12 min.
- The programme is managed from Bremen, where development, integration and testing of the experiment facilities take place.

Airbus Defence and Space

Air Power

AIRBUS

A400M



Bremen is a vital part of Airbus‘ A400M value chain:

- Integrated fuselage structural development and general aircraft systems installations design.
- Cargo hold interiors and aerial delivery systems development.
- End-to-end steering of major suppliers and external and internal partners activities.

Cargo Hold System



- An A400M can **transport** numerous pieces of outsize cargo including vehicles and helicopters.
- Maximum payload: Up to 37 tonnes, volume: 340 m³
- Bremen's responsibilities:
  - Cargo and Aerial Delivery System Engineering, Operation and Procedures
  - Functional testing
  - Loadmaster software development

- An A400M provides **aerial delivery** capabilities of payload and paratroopers by parachute automatic to calculated points.
- Maximum airdrop weight: up to 25 tonnes or 115 paratroopers.
- Bremen's responsibilities:
  - Functional testing
  - System development
  - Certification



A400M Integrated Fuselage Assembly

- Center and rear fuselage components & assembly
- Cargo door
- Hydraulic, electrical and mechanical systems
- Functional test
- Support customer inspection on site or inspection on behalf of the customer
- Final Assembly Line support for A400M fuselage activities

AIRBUS

Future Combat Air System (FCAS)



- A key project in ensuring European autonomy and sovereignty in defence and security.
- FCAS will be centred around a core Next Generation Weapon System (NGWS). In this “system of systems,” piloted New Generation Fighters (NGF) working together with Unmanned Remote Carriers (RC) – all interconnected with other systems in space, in the air, on the ground, at sea and in cyberspace using the “Combat Cloud” (CC) as common infrastructure.
- Bremen contributes to NGWS and FCAS in the following areas:
  - System of System, Next Generation Fighter, Remote Carrier, Enhanced Low Observability Technology.

Airbus Defence and Space

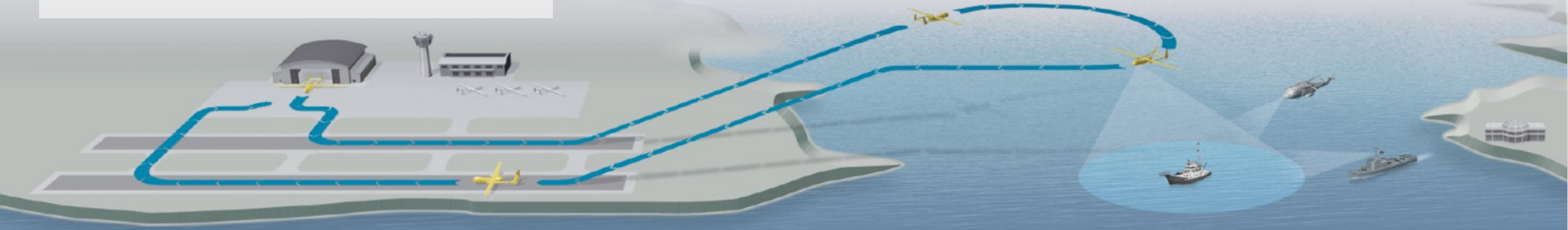
Airbus DS Airborne Solutions

AIRBUS

Unmanned Aerial System (UAS) Operational Services

Consulting
- Requirement analysis and system design
- Supporting bid and contract documents
- Supplier management

Operations
- Worldwide UAS operation and support
- Co-operative system support centre
- Lease-operate-maintain service
- Service for customer-owned systems



Training
- Maintenance personnel
- Payload operator
- Air vehicle operator (unmanned aerial vehicle pilot)

Support
- System qualification and certification
- Initial and continuing airworthiness
- System implementation
- Integrated logistical support (documentation, spares/repairs, maintenance, …)

AIRBUS



UAS Ground Control Stations

Ground Control Stations (GCS) for Mission Management, Payload-Management, Exploitation Command and Control Interface

- Highly flexible GCS concept
- Competencies along customer needs
- Data driven scalability of customer interface

AIRBUS

Cargo Loading Systems (CLS)



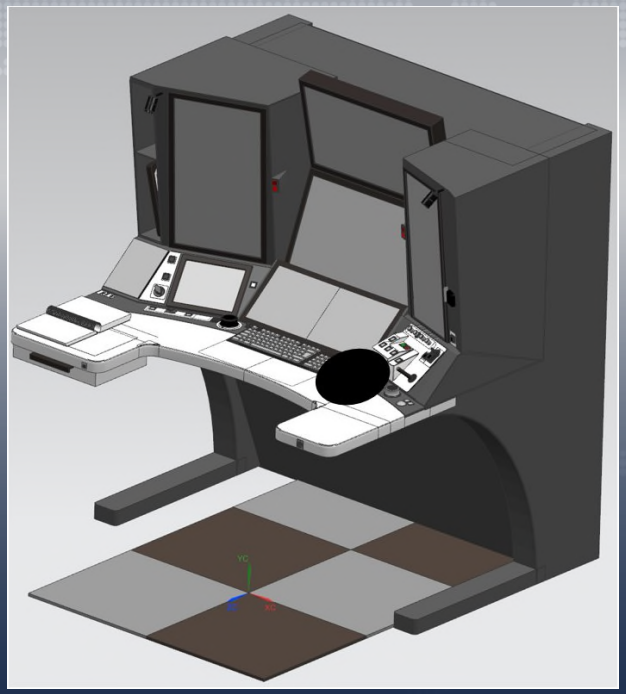


CLS experience built on years of delivering innovation

- Cargo loading systems for passenger, freighter and military aircraft
- Load master controls and pax to freighter conversion
- Production and in-service support
- Repair and overhaul

AIRBUS

Engineering & Tool Chain



Main engineering assets available to drive concept and design phase and to ensure continued airworthiness

- Mechanical and electrical installation
- Structures/airframe
- General air vehicle and general systems
- Mission systems (blue chain)
- Test
- Research & Technology/Research & Development
- Avionics and payload integration
- Production support/continued airworthiness
- UAS certification and services
- UAS ground control stations/human machine interface (HMI)
- Cargo load controls

Production & Maintenance



Assets available to enable stable and reliable production and maintenance processes

- POA/MOA by EASA and GMAA
- Engineering support
- Mechanical and electrical installation
- Avionics
- Maintenance
- Structure repairs
- Non-destructive testing
- Continuing airworthiness management organisation (CAMO) tracking

ArianeGroup

ArianeGroup



Airbus Amber

- ArianeGroup guarantees Europe‘s independent access to space
- Main contractor for Ariane 6 launchers and developer of future upper stages
- Supplier of highly specialized propellant tanks for satellites and orbital vehicles
- Supplier for the Orion European Service Module
- 1250 employees in Germany incl. more than 600 in Bremen

arianeGROUP

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Ariane 6

[Airbus Amber]



Development, industrialization and integration of the new Ariane 6 upper stage (Upper Liquid Propulsion Module)

[Airbus Amber]

Astris kick stage



- Development of the Astris kick stage, an optional add-on to Ariane 6's upper stage
- Enables Ariane 6 to offer a range of new space transportation services by allowing complex orbital transfers
- Astris will simplify missions by taking over some of the required built-in propulsion capabilities of payloads to move themselves to their final position in orbit

arianeGROUP



[Airbus Amber]

Tanks & Components

Development and production of propellant tanks for satellites and orbital vehicles

ArianeGroup

Testia

TESTIA
AN AIRBUS COMPANY

NDT Inspection & Training



- All non-destructive testing methods
- Inspection for manufacturing and in-service (Part-145)
- NDT Training Center

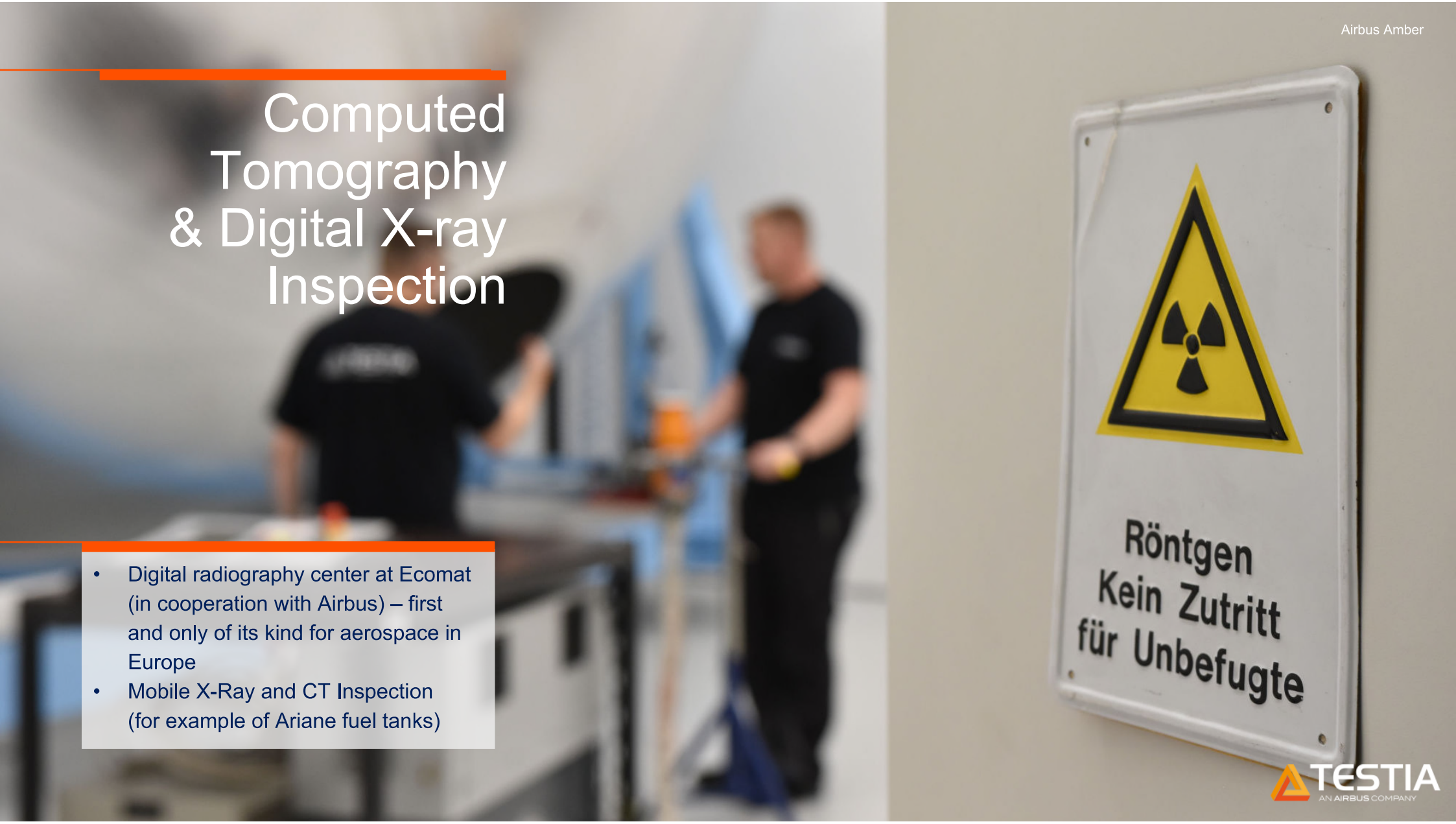
TESTIA
AN AIRBUS COMPANY



Materials & Processes

- > 100 material tests available
- sample manufacturing
- Test lab for aircraft interiors
- Failure analysis
- Qualification support for Airbus suppliers

TESTIA
AN AIRBUS COMPANY



Computed Tomography & Digital X-ray Inspection

- Digital radiography center at Ecomat (in cooperation with Airbus) – first and only of its kind for aerospace in Europe
- Mobile X-Ray and CT Inspection (for example of Ariane fuel tanks)

TESTIA
AN AIRBUS COMPANY

Structural Health Monitoring (SHM)



- Testia is an end-to-end SHM system integrator, from planning to installation
- wide range of sensor types
- cloud-based multi-sensor technology
- SHM data analytics service
- consultancy

TESTIA
AN AIRBUS COMPANY

3D Metrology



Airbus Amber

- Testia is a certified provider of GOM Metrology
- Automated scanning of parts, jigs and tools
- Reverse engineering of 3D models based on scan data

TESTIA
AN AIRBUS COMPANY

Cross-company activities to secure the future

AIRBUS

Centre for Eco-Efficient Materials & Technologies



Research activities:

- Hydrogen technologies for aviation (ZEROe Development Centre)
- Future intelligent wing concepts
- Sustainable materials & processes (e.g. chromate-free primer, ALM/3D printing, cryogenic materials, recycling)
- Virtual product engineering

- Home of all Research & Technology related activities and materials & technologies research
- Efficiency through collaboration with industrial key partners and universities

AIRBUS

ZEROe Development Centre (ZEDC)

- **Leveraging LH2 tank core technology** (systems installation, cryogenic and ambient testing)
- **Fast make activities** and relevant **ground tests/flight tests demonstrations**
- Building on **LH2 experience** at Airbus Defence and Space and ArianeGroup
- Progressively transfer competences to other sites using **chamber of reference model**
- Key requirement of ZEDC: **resilience** to changes in tank architectures and technology



AIRBUS

Vocational Training



- **One Vocational Training** for all divisions on site
- Various Apprenticeships and Dual Study Programmes
- Airbus Graduate Qualification Programme (AAF)
- International Experiences / Mobility for Airbus Apprentices (M4AA)
- Focus on digitisation, automation and future technologies

Figures

- **220** apprentices and dual students working for Commercial Aircraft, Defence and Space, Airbus Aerostructures and Ariane Group

AIRBUS

Supporting our business

- Our teams e.g. from Environment, Health & Safety, Facility Management, Human Resources, Procurement and Security enable, support and protect all activities on site



Thank you

© Airbus

AIRBUS

Core competencies at Bremen site

Airbus Operations



Materials & Processes



Flight physics



High-lift systems E2E
(Design, Testing, Assembly, Repair)



Hydraulic Systems Engineering



FAF Engineering
ESI, MSI, Structure, Doors, Stress & Repair Engineering



Cabin & cargo system engineering



European Repair Station



ZEROe Development Centre



Wing Equipping A330/A350

Airbus Aerostructures



Production and technology development of sheet metal and CFRP formed parts

Ariane Group



Development and Assembly of Upper Stages for Ariane launchers

Airbus Defence and Space: A400M & Space Systems



A400M fuselage & cargo development and assembly
Development of military capabilities



Space Exploration, Human space-flight & Space Station System Engineering

Airbus Defence and Space: Airbus DS Airborne Solutions



UAS Airspace Integration/ Registration/ Certification
UAS Services

AIRBUS

Kernkompetenzen am Standort Bremen

Airbus Operations



Materialien, Prozesse & Testing



Flugphysik



Hochauftriebssysteme E2E
(Design, Testing, Montage, Repair)



Hydraulik System Engineering



FAF Engineering
ESI, MSI, Structure, Doors, Stress & Repair Engineering



Kabine & Frachtsystem Engineering



ZEROe Development Centre



Flügelausrüstung A330/A350



European Repair Station

Airbus Aerostructures



Produktion & Technologie-entwicklung von Metall- und CFK-Umformteilen

Ariane Group



Entwicklung & Fertigung von Oberstufen für Ariane Trägerraketen

Airbus Defence and Space: A400M & Space Systems



Entwicklung / Montage A400M Rumpf und Frachtsystem
Entwicklung militärischer Fähigkeiten



Astronautische Raumfahrt
Raumfahrt-Systemtechnik
Entwicklung Raumstation

Airbus Defence and Space: Airbus DS Airborne Solutions



UAS Luftraum Integration/ Registrierung/ Zertifizierung
UAS Services

AIRBUS